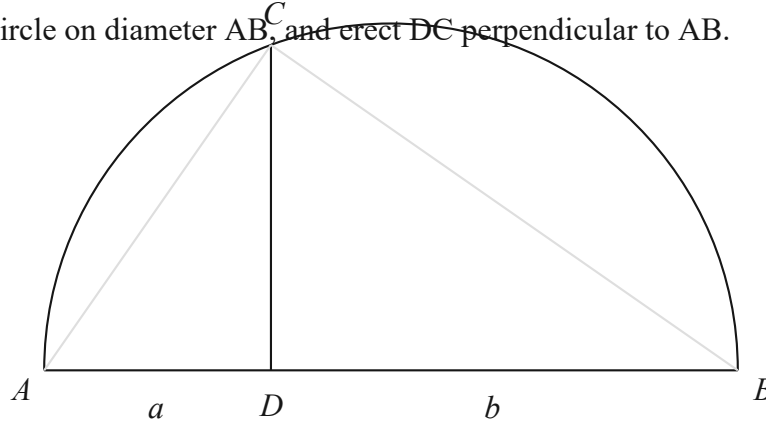


01 / A VISUAL PROOF

Geometric inequalities and proofs

A semicircle gives a picture of the arithmetic-geometric mean inequality. Let $AD = a > 0$ and $DB = b > 0$. Draw the semicircle on diameter AB , and erect DC perpendicular to AB .



The angle ACB is right. Similarity of the two smaller right triangles gives $CD^2 = AD \cdot DB = ab$. Thus CD is the geometric mean of a and b .

The radius is $R = (a + b)/2$. If O is the midpoint of AB , the right triangle ODC gives $CD^2 = R^2 - OD^2$, so CD cannot exceed R .

$$\sqrt{ab} \leq (a + b)/2, \text{ with equality exactly when } a = b.$$

Three inequalities to keep nearby

1. Arithmetic-geometric mean. For $a, b \geq 0$,

$$(a + b)^2 \geq 4ab.$$

Proof: subtract $4ab$ from the left to obtain $(a - b)^2 \geq 0$. Equality holds exactly when $a = b$. For positive variables, taking square roots gives the familiar mean inequality.

2. Triangle inequality for lengths. The side lengths a, b, c of a non-degenerate triangle satisfy $a + b > c$, with analogous inequalities after cycling the sides.

Proof: the broken path along two sides is longer than the straight segment joining its endpoints. More algebraically, the cosine law gives $c^2 = a^2 + b^2 - 2ab \cos C < (a + b)^2$, since $0 < C < 180$ degrees.

3. Two-dimensional Cauchy-Schwarz. For real x, y, u, v ,

$$(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2).$$

Proof: the right-hand side minus the left-hand side equals $(xv - yu)^2$, which is nonnegative. Equality holds exactly when $xv = yu$, meaning the two vectors are linearly dependent.

The common theme is to make the difference between two sides visibly nonnegative. A square, a length, or an area is often the right object.

Optimisation without calculus

Worked example. A rectangle has perimeter 20. If its side lengths are a and b , then $a + b = 10$. AM-GM gives $ab \leq 25$, attained exactly at $a = b = 5$. The square has the largest area.

Exercises

1. A rectangle has area 36. What is its smallest possible perimeter?
2. If $x^2 + y^2 = 1$, what is the largest possible value of $3x + 4y$?
3. Two sides of a triangle have lengths 5 and 8. What open interval contains its third side c ?

Solutions

1. AM-GM gives $a + b \geq 2\sqrt{36} = 12$. The perimeter is at least 24, attained by the 6 by 6 square.
2. Cauchy-Schwarz gives $(3x + 4y)^2 \leq 25$, so $3x + 4y \leq 5$. Equality is attained at $(x,y) = (3/5, 4/5)$.
3. The triangle inequalities give $c < 13$ and $c > 8 - 5 = 3$. Hence $3 < c < 13$. Both endpoints would make a degenerate triangle.

When solving an optimisation problem, always state both the bound and a configuration that attains it. Without the second part, you may only have an estimate.